

Kitchcu

Complete Executive Guide - CEO, CPO & CTO

Kitchcu cloud kitchen platform

Audience: CEO, CPO, CTO, Product, Engineering, Investors

Three executive lenses:

- CEO - positioning, subscription economics, risks, current platform scale
- CPO - every module defined with challenges solved + journeys + KPIs
- CTO - architecture, event/data flows, logical ER, APIs, security

This guide includes:

- 18 product modules (Gateway through Live Stream + GST + Quality Loop design)
- Sprints S1-S18 shipped: microservices, PWAs, billing, GST, ratings, growth
- Challenge map: owner P1-P12 and customer C1-C6 mapped to modules
- CTO diagrams: system architecture, order/payment/GST/quality flows, ER
- Zero per-order food commission; owner-owned CRM; live-capture truth
- Next: E1 purchase inventory + E2 chef standard lock (design pack ready)

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Executive Strategy

Kitchcu is a B2B2C cloud kitchen operating system - not a food aggregator and not a restaurant POS. Owners run orders, menu, quality, CRM, payments, GST, and growth from one PWA. Customers get live-capture honesty, fair fees, home-taste ratings, and multi-kitchen checkout without surrendering ownership or paying per-order commission.

Stakeholder	Challenge	Kitchcu answer
Owner/chef	WhatsApp chaos, commissions	Unified hub + subscription SaaS
Customer	Fake photos, opaque fees	Live media + fee quotes + tracking
Platform	Capital-efficient SaaS	PWA-first event-driven services

"Keep day-one simple: accept an order and see revenue in minutes. Growth layers unlock after traction."

S1-S18

Sprints shipped

13

Domain services

Live

GST finance

Design

E1/E2 quality loop

2. Market Positioning & Business Model

Aggregators (marketplace)	Kitchcu (operating system)
----- Per-order commission 18-30% Platform owns the customer Stock / studio photos Speed race delivery timers Single kitchen cart No GST / quality OS	----- Flat subscription Owner owns CRM + data Live-capture dish media Owner-set quality SLA Multi-kitchen master checkout GST + ingredients + standards

Non-negotiable: zero per-order food commission. Kitchen keeps food revenue.

Tier	Monthly	Role
Starter	Rs 499	Operations + basic reports
Growth	Rs 999	CRM, coupons, deeper insight
Pro	Rs 1,999	Multi-kitchen, priority support

Metric	Year 1 target
CAC (owner)	< Rs 2,000
Owner LTV	> Rs 18,000
LTV:CAC	> 3:1
Gross margin	> 75% SaaS

3. Go-to-Market Phases & Risks

Phase	Goal	Status
1 Foundation	Owner runs kitchen end-to-end	COMPLETE S1-S18
2 Growth polish	Offline PWA, CRM automation	Continuous
3 Quality loop	Purchases + locked chef standards	Design pack
4 Scale	National / white-label	Future

Risk	Mitigation
WhatsApp API change	Manual + PWA intake always on
Scope creep	Design pack gate before code
Rating fraud	Verified purchase only
Multi-kitchen refunds	Sub-orders + Route settlements
Onboarding friction	5-min kitchen -> dish -> order

Product Modules & Challenges

Every cloud kitchen scales like a brand - with data, taste standards, and direct customers - while diners see honest, real-time food truth.

Persona	Promise	Surface
Owner / Chef	WhatsApp -> revenue same day	kitchen.kitchcu.in
Customer	Trust, fair delivery, rate taste	customer.kitchcu.in
Admin	Support + attention queue	admin.kitchcu.in
Market visitor	Brand story & signup	portal.kitchcu.in

- Quality over speed - owner-set prep/delivery windows
- Truth in media - live-capture heroes; no stock photo deception
- Owner owns CRM - spend, coupons, patterns stay with kitchen
- Progressive complexity - hide advanced until traction
- Growth OS, not restaurant POS / dine-in / KDS

2. Challenges to Module Solutions

Owner challenges (P1-P12)

ID	Challenge	Modules
P1	Orders trapped in WhatsApp/calls	Order + Notification
P2	Aggregator commission kills margin	Billing subscription
P3	No daily revenue / segments	Analytics + Growth + Reports
P4	Stock photos erode trust	Catalog live-capture
P5	Taste drifts batch to batch	Recipes + Ratings + E2
P6	Customer data owned by platforms	Marketing CRM
P7	Promotions are guesswork	Growth + daily menu
P8	Multi-channel lifecycle chaos	Order status machine
P9	Stock unknown mid-service	Ingredients + E1 purchases
P10	GST / monthly audit pain	Billing GST
P11	Skills & trials hard to run	Learning + Community
P12	Trust without ad spend	Streaming live opt-in

Customer challenges (C1-C6)

ID	Challenge	Modules
C1	Cannot trust menu photos	Catalog
C2	Opaque delivery fees	Delivery
C3	Weak tracking	Order + Notify + Delivery
C4	Generic star ratings	Ratings home-taste
C5	One kitchen per cart	Master checkout + Billing
C6	Hard to re-find local kitchens	Identity nearby discovery

3. Module Catalog - Definitions & Challenges

Each module is a bounded product context. Definition = what it is. Description = what it does. Challenge = kitchen/customer pain it removes.

M01 API Gateway

- Definition: Public edge for /api/v1.
- Description: Path proxy, CORS, correlation IDs; no business logic.

- Challenges solved: Clients never hit internals; unified auth surface.

M02 Identity & Kitchen

- Definition: Auth + tenant roots.
- Description: Owner OTP/JWT; customer OAuth/OTP; PostGIS kitchens; codes CKPNQ001.
- Challenges solved: Onboarding bootstrap; nearby discovery; multi-kitchen ownership.

M03 Catalog & Live Media

- Definition: Menu of truth.
- Description: Categories, dishes, prices; hero must be live-capture; menu cache.
- Challenges solved: Ends stock-photo deception; consistent menus across PWAs.

M04 Ingredients & Recipes

- Definition: Pantry + dish standards (F19).
- Description: Stock, thresholds, recipe lines, prep steps; deduct on accept.
- Challenges solved: Stops mid-service stock outs; foundation for taste standards.

M05 Order Operations

- Definition: Intake, lifecycle, bills, analytics.
- Description: Manual/WA/PWA; status machine; PDF bills; revenue/peak/segments.
- Challenges solved: Single operational truth for chaotic multi-channel demand.

M06 Multi-Kitchen Checkout

- Definition: One cart, many kitchens (F06).
- Description: Master order + atomic sub-orders + master receipt PDF.
- Challenges solved: Removes aggregator-style one-kitchen cart limit without commission.

M07 Billing & Settlements

- Definition: Money movement.
- Description: Payments, UPI, subscriptions; Route splits (F44).
- Challenges solved: Subscription replaces commission; fair multi-kitchen payouts.

M08 GST Finance

- Definition: Tax profile, invoices, monthly audit.
- Description: GSTIN profiles; sync from delivered orders; balance sheet close.
- Challenges solved: Accountant-ready monthly GST for home/cloud kitchens.

M09 Notification & Support

- Definition: WhatsApp + tickets + tracking nudges.
- Description: Webhook intake; F45 updates; F29 reminders; AI support tickets.
- Challenges solved: Closes communication gaps owners cannot staff manually.

M10 Marketing & CRM

- Definition: Owner-owned relationships.
- Description: Customer spend history, coupons, targeted promotions.
- Challenges solved: Recaptures customers aggregators monopolized.

M11 Ratings & Tips

- Definition: Verified home-taste + F20 tips.
- Description: 0.6 taste + 0.4 quality; A/V reviews; owner tip workflow.
- Challenges solved: Quality benchmark customers believe; fuel for chef briefs.

M12 Growth Intelligence

- Definition: Actionable suggestions, not vanity charts.
- Description: Combos, patterns, grow cards, daily menu WhatsApp push.

- Challenges solved: Ends promotion guesswork with evidence-backed actions.

M13 Delivery & Tracking

- Definition: Distance fees + shareable track.
- Description: PostGIS quotes; free/max radius; tracking tokens/links.
- Challenges solved: Transparent fees and journey visibility.

M14 Learning & Trials

- Definition: Skills + controlled experiments.
- Description: Curated learning; dish trials with promote-to-menu path.
- Challenges solved: Helps home chefs professionalize without outside help.

M15 Community & Rankings

- Definition: Recipes + chef leagues.
- Description: Recipe rewards; rankings from quality/activity signals.
- Challenges solved: Differentiation and motivation vs pure marketplaces.

M16 Live Streaming

- Definition: Owner opt-in LiveKit sessions.
- Description: Go-live controls; customer live kitchen filter.
- Challenges solved: Trust and engagement without paid advertising.

M17 Website PWAs

- Definition: Installable surfaces for all personas.
- Description: React+Vite; Workbox; owner command-center UX; maps; RTE.
- Challenges solved: App-store-free distribution; WhatsApp-native deep links.

M18 Quality Loop E1+E2

- Definition: Purchases + lock winning standards.
- Description: Purchase ledger stock-in; chef brief from volume/ratings; lock.
- Challenges solved: Closes taste drift + pantry truth as one product loop.

4. Product Journeys & Capability Ladder

Owner day-1

```
Register -> OTP -> JWT -> Create kitchen (geo + code)
-> Add dish (live hero) -> Optional GST profile
-> First manual / WhatsApp order -> Accept -> Deduct stock
-> Same-day revenue report
```

Customer trust -> rate

```
Nearby map -> Menu (live photos) -> Fee quote -> Checkout
-> Pay (single or multi-kitchen) -> Track -> Delivered
-> Home-taste rating (+ optional tip)
```

Capability ladder (progressive complexity)

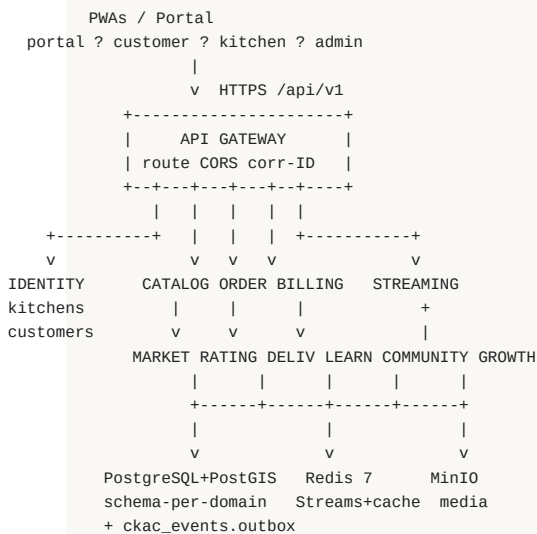
- 1. Menu + Orders + Reports
- 2. CRM + Coupons + Delivery quotes
- 3. Ingredients + GST + Growth suggestions
- 4. Learning + Community + Live stream
- 5. Quality loop lock (E1 purchases + E2 chef standards)

5. Product KPIs

KPI	Near-term	12-month
Active kitchens	10 pilots	500
Platform orders/day	50	5,000
Owner monthly retention	80%	90%
30-day customer repeat	25%	40%
Locked standards / kitchen	-	>= 3 post-E2
GST audits closed on time	-	>= 80% GST kitchens

Architecture, Flows & Data

Rule: one bounded context per container; cross-service writes via events; cross-schema reads only for ownership/evidence; gateway is sole public edge.



2. Event & Data Flow Diagrams

Order + stock + notify

```

Owner/Customer place|accept
|
v
ORDER SERVICE
- persist order + status_events
- publish order.placed / order.status.changed
- on accept --internal HTTP--> CATALOG deduct
                                publish stock.deducted
|
v Redis ckac:orders:order
+--> NOTIFICATION (WhatsApp F45, tracking nudge)
  
```

Multi-kitchen payment (F06/F44)

```

Cart [K1, K2] -> ORDER master_order + sub-orders
-> BILLING master payment capture
-> settlements[] net_to_owner per kitchen
-> events payment.captured / settlement.*
-> master bill PDF for customer
  
```

GST monthly loop

```

GSTIN profile -> kitchen_gst_profiles
Delivered orders --sync--> gst_tax_invoices
Monthly report + balance sheet
Close audit -> gst_monthly_audits snapshot
  
```

Quality loop E1/E2 (planned)

Purchase post -> stock_movements(+) -> pantry
 Accept order -> stock_movements(-) via recipe
 Ratings+volume+F20 tips -> GROWTH chef_brief
 Owner Lock -> recipe_standard_versions + dish_ingredients

Representative Redis streams

Stream	Example events
ckac:orders:order	placed, status.changed
ckac:catalog:*	dish/ingredient stock.*
ckac:billing:gst	profile, invoice, audit
ckac:ratings:*	created, aggregate.updated
ckac:growth:suggestion	generated (+ chef_standard)
ckac:notify:*	whatsapp, tracking

Every write publish uses transactional outbox (ckac_events.outbox).

3. Logical ER / Schema Diagram

Schemas are separate Postgres schemas (no cross-schema FKs). Lines below are logical relationships only. Tenant tables carry kitchen_id.

ckac_identity	ckac_catalog	ckac_orders
-----	-----	-----
owners 1--* kitchens	categories 1--* dishes	master_orders
customers	dishes 1--* dish_media	1--* orders
	dishes *--* ingredients	orders 1--*
	dish_prep_steps	order_items
		status_events
ckac_billing	ckac_ratings	ckac_marketing
-----	-----	-----
subscriptions	dish_ratings	kitchen_customers
payments 1--* settlements	dish_rating_aggregates	coupons
gst_profiles/invoices	dish_suggestions	promotions
gst_monthly_audits		
ckac_growth delivery learning community streaming		ckac_events
suggestions quotes* trials rewards sessions		outbox
seasonal_* track* lessons rankings		processed

4. Services, APIs, Security & Standards

Service	Port	Highlights
gateway	18000	/api/v1/* edge
identity	18001	auth, kitchens, customers
catalog	18002	menu, dishes, ingredients
order	18003	orders, analytics, PDF
billing	18004	payments, GST
notification	18005	webhooks, support
marketing	18006	CRM, coupons, promos
ratings	18007	ratings, suggestions
growth	18008	suggestions, daily menu
delivery	18009	fees, tracking
learning	18010	portal, trials
community	18011	recipes, rankings
streaming	18012	LiveKit sessions

- Security: JWT Bearer, tenant filters, Pydantic, env secrets, mask PII, X-Internal-Key
- Health: /health/live + /health/ready on every service
- Method: TDD + EDD; MODULE-DESIGN-PACK before new features; Alembic only

- Forbidden: per-order commission, restaurant POS/dine-in/KDS, cross-schema writes

5. Build Status Matrix

Module band	Sprint	Status
Gateway / Identity / Catalog / Order / N	S1-S4	DONE
PWAs + checkout + analytics	S5	DONE
Billing + GST	S6+GST	DONE
Discovery / history	S7	DONE
Multi-kitchen cart	S8	DONE
Split payment Route	S9	DONE
CRM / coupons / promos	S10	DONE
Ratings	S11	DONE
Growth	S12	DONE
Delivery	S13	DONE
Tracking notify	S14	DONE
Ingredients	S15	DONE
Learning / Community / Streaming	S16-S18	DONE
E1 Purchases + E2 Chef lock	S19	DESIGN

Reference

Band	Features	Status
Orders & lifecycle	F01-F05, F30, F45	Done / WA AI partial
Multi-kitchen & pay	F06, F42-F44	Done
Analytics & growth	F07-F12, F39	Done
Catalog & media	F13-F15	Done
Ratings	F16-F18, F20	Done (F20 UI thin)
Ingredients	F19	Done (E1 extends)
Delivery / discover	F27-F33	Done
Marketing	F34-F41	Core done
Learning / community	F21-F24	Done
Live	F46-F48	Done
GST finance	Billing extension	Done
Quality loop	E1/E2	Design pack

Full acceptance criteria: docs/CKAC-COMPLETE-PLANNING-BENCHMARK.md

2. Document index

Document	Role
CKAC-COMPLETE-GUIDE.md/.pdf	This master CEO/CPO/CTO guide
E1-E2-*~DESIGN.md	Next sprint quality-loop design
CKAC-IMPLEMENTATION-GUIDE.md	Built features mapped to code
KITCHCU-ENGINEERING-STANDARDS.md	Engineering constitution
MODULE-DESIGN-PACK.md	Mandatory pre-code template
AGENTS.md	Agent/engineer quick spec
Feature packs F*.md	Per-sprint design packs

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